

F24-081-D-newsNexus

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Chapter 1

Introduction

The purpose of this report document is to present a detail plan for the development of newsNexus. The report discusses the objectives, technical requirements, and much more with a focus on improved news delivery to users, ease of access, and overall user engagement.

The newsNexus will enhance the already existing weaknesses revolving around news consumption through artificial intelligence. It will customize news for individual users at a finely tailored and optimized level. In its essence, newsNexus solves the generic problems of news consumers such as the inability to search for specific details within the news articles, comprehend complicated subjects, and receive news according to one's needs and previous consumption patterns. It uses the information which is associated with each user and changes with time, synthesizing up-to-date information, brief notes on articles and many views on different issues.

Thanks to the way newsNexus offers news, it is totally a new and improved world. It is not just a news platform, it's a gateway to a more informed and engaged world. By harnessing the power of AI, we aim to streamline the news consumption process, making it more efficient and enjoyable while fostering a deeper understanding of global events.

1.1 Problem Statement

In today's digital world, the available news are so much that the users often do not digest the information efficiently. It can be time consuming and annoying to search for any specific information within the news articles, especially when there are topics that are difficult to understand and relate. Even though there are many news sources available, but the readers are not shown news that aligns with their past interactions, searches, or preferences which leads to lack of personalization in the news consumption process. Thus, users often waste time on irrelevant content that they are not passionate about, and struggling to find what they truly need. Moreover, current news platforms don't have features with which users can save articles they find

interesting or want to read again. Due to this, users have to repeatedly search for the same content which wastes a lot of time and effort. When trying to compare perspectives on a single topic, users often have to manually sift through multiple articles, and read them all to get an idea where the difference is or what the difference is. This lack of personalization, with the inability to easily save and compare content, tells about the inefficiencies in the current news consumption process, which highlights the need for a more tailored and user friendly solution.

1.2 Problem Solution

To address the challenges faced by users in navigating today's overwhelming news landscape, newsNexus provides a **fresh and user-centered approach** to news consumption. The platform **offers a personalized news feed** that adapts to each user's past interactions, search habits, and specific preferences. This ensures that readers see the most relevant content, saving time and reducing frustration from sifting through unwanted news.

With newsNexus, articles are provided in **concise, easy-to-read summaries alongside real-time updates**, making it simpler for users to stay informed without feeling overwhelmed. The platform also **tackles the issue of perspective diversity** by gathering content from multiple trusted sources on the same topic, allowing users to effortlessly explore different viewpoints without manually searching through countless articles.

Furthermore, newsNexus incorporates advanced features, such as a powerful **search function, topic-level predictions, and intelligent user profiling**, to create a more intuitive and engaging news experience. Users can type or record queries, get detailed answers from articles or video clips, and save responses they find interesting for future reference—all in one place.

Ultimately, newsNexus turns the often tedious task of finding, reading, and comparing news into a **streamlined and tailored experience**. It helps users stay informed in a way that **fits their unique interests** while enhancing their ability to save and compare content, making the news consumption process more efficient and enjoyable.

1.3 Scope

1.3.0.1 Overview

newsNexus is a personalized news aggregation platform designed to collect, process, and present news articles to users based on their preferences. It provides a tailored experience by allowing users to select their preferred news topics and sources, along with a query feature for deeper insights, including video content and future predictions.

1.3.0.2 Objectives

- To provide users with personalized news content tailored to their interests.
- To enable users to interact with the news through text or voice commands.
- To incorporate multiple types of media, including text articles and video clips, in response to user queries.
- To offer predictive insights into topics of interest using AI.

1.3.0.3 Key Features

User Registration and Login: Users can create an account by providing their email, username, and password. Returning users can log in using their credentials.

Personalization: Users are required to select their preferred news categories (e.g., sports, health, politics) during initial setup. These preferences drive the content displayed in the news-Nexus interface.

News Interface: Once logged in, users are presented with a customized news feed displaying articles based on their preferences. The interface is designed for easy navigation, allowing users to browse articles relevant to their interests.

Query Functionality: Users can type or record their queries, which can range from basic news searches to requests for detailed explanations or future predictions.

Examples of supported queries include:

- Specific news information (e.g., “Information about electric buses in Multan”).
- Statements made by public figures (e.g., “What did Hamid Mir say in his talk show on September 20, 2024?”).
- Topic-based queries (e.g., “Show articles related to renewable energy”).
- Requests to simplify a complex article.
- Predictive queries (e.g., “What will the job market be like in Pakistan in the coming years?”).

Comprehensive Responses:

- Responses include article references, news sources, dates, and times.
- Users can receive relevant information from YouTube, including video clips that specifically address the query.

Saved Responses and History: Users can save responses for future reference and view their saved items in history pane. This feature helps users keep track of important information they might need later.

Logout Functionality: Users can securely log out of their account when finished.

1.3.0.4 Boundaries and Exclusions

- The platform will only scrape data from five defined news sources.
- The system does not permanently store YouTube content; it temporarily processes and delivers clips based on user queries.

1.3.0.5 Deliverables

- A fully functional news aggregation platform with user accounts, preferences, and query handling.
- An intuitive user interface that allows for both text-based and voice commands.
- Integration with multiple data sources, including text and video, to deliver relevant news content.
- A backend system capable of storing and managing user data, including preferences, saved queries, and search history.

1.3.0.6 Assumptions and Constraints

- **Assumptions:** Users have internet access for content retrieval. News sources allow periodic scraping of content.
- **Constraints:** The system relies on third-party news sites and YouTube APIs, which may introduce data access limitations. User queries are processed within the bounds of the available data; predictions are limited by historical data quality.

1.3.0.7 Success Criteria

Users can successfully register, log in, set preferences, and receive personalized news content. Users can use both text and voice queries to search for information. The platform provides accurate, up-to-date responses to user queries, including predictions and video segments when applicable.

1.3.0.8 User Journey

When users open newsNexus, they create an **account** in it. They enter email, user name and password. After the login and sign up process, users are required to select their **news preferences** such as sports, health or politics and after that users are displayed with the interface of newsNexus where they can see the articles according to their preferences. If user have queries regarding any news article or want to get information regarding available news or topic level prediction, they can **search** their relevant query and for a hand free experience they can also **record** their queries. The query can be anything related to the news or articles available that is

- Any information about electric buses in Multan?
- What did Hamid mir said in talk show of 20 September 2024?and also tell the source
- News about why did rota virus new vaccine was not beneficial!
- Show articles related to renewable energy!
- Cannot understand this article, make it easy!
- What will the job market be like in Pakistan in few years?

And much more. With this the user can also get answer according to their preferred news sources. The answer to the queries will give date, time, article reference and news source with related information needed.The user can also get answer from a YouTube video and the clip related to the answer. The user can **save** the response they might need for future use and can also **view** the saved responses and history. After that if user wants to **logout** they can.

1.4 Modules

1.4.1 Personalization Module

This module is responsible for delivering a tailored news experience based on users individual preferences. It basically tracks and learns from user behavior to refine the recommendations and improve content relevance for the users.

1. **Personalized News Feed:** Tailors the news content based on individual user preferences and interests. The system continuously learns from user interactions to deliver increasingly relevant news.
2. **User Profiling:** Develops the profile for each user based on their reading habits and preferences, improving the accuracy of content personalization.

1.4.2 News Content Management Module

This module manages the flow of news which includes real-time updates, article summaries and news categorization. This part make sure that news is presented efficiently and in an organized manner for easy user navigation.

1. **Real-Time News Updates:** Provides news updates, ensuring users have access to the latest news as events unfold.
2. **Article Summaries:** Generates concise summaries of lengthy articles, allowing users to quickly grasp the main points without reading the full text.
3. **News Categorization:** Organizes news articles into categories such as politics, sports, technology etc. which helps the users to easily navigate through various news topics.

1.4.3 Source Management Module

In this module, it will allow the users to select based on their preferences if they want.

1. **Source Selection:** Allows users to choose their preferred news sources.

1.4.4 Search and Query Module

This module allows users to search for news articles and topics or any news related query they want. The query can be in the form of text and voice search and also provides video responses for certain queries.

1. **Search Functionality:** Enables users to search for specific news articles or topics using text-based queries.
2. **Audio Integration for Voice Queries:** Lets users search for news using voice commands which is another way of improving accessibility.
3. **Query Response by video:** When a user asks a query related to a video (e.g., "Hamid Mir about US air bases in Pakitan" or "what did Aftab Iqbal said in todays show"), the system will provide a text-based reply and also return a specific clipped video relevant to the query.

1.5 User Classes and Characteristics

The user classes of my project will be the user but below I have categorized what type of user will be using News Nexus.

User class	Description
Casual News Reader	Description: Users who access news for general awareness, entertainment, or to keep up with trending topics. They may use the platform occasionally or daily but are not deeply invested in a specific topic. Characteristics: Interested in personalized news feeds, real-time updates, article summaries, and easy-to-navigate categories. They appreciate quick and concise information and may use audio integration or voice commands for a more interactive experience.
Informed User/Researchers	Description: Users who actively seek in-depth knowledge about specific topics, such as researchers, journalists, or students. They rely on newsNexus to access detailed, comprehensive news and multiple perspectives. Characteristics: Use advanced search functionality, source selection, and video query responses for a thorough understanding of topics. They appreciate user profiling that aligns with their evolving interests and preferences, enabling them to access diverse viewpoints and more profound insights.
Content Creators/Analysts	Description: Bloggers, journalists, or analysts who utilize the platform to gather news insights, identify trends, or analyze multiple viewpoints. They may also contribute by providing feedback on the accuracy of AI-generated summaries or content. Characteristics: Require access to diverse sources, category-specific news, and features like video query responses to gather comprehensive content. They might use the search and audio integration features to enhance their research or article creation process.
Project Managers and Stakeholder	Project managers and stakeholders oversee the development and progress of newsNexus, ensuring the platform aligns with the project's objectives. They track how the application is working.

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use Case



Figure 2.1: Use Case Diagram

The use case diagram for the News Nexus system illustrates the interactions between two primary actors: the User and the System.

- **User Actions:** The User can perform several actions, including signing up, logging in, selecting preferences, choosing news sources, reviewing their history, typing queries, saving responses, and logging out. The user can type their query either as text or by recording audio.
- **System Responses:** Once the User logs in, the system verifies the username and password. If the credentials are incorrect, an error message is displayed. The system also extends other use cases such as processing the user's request to get an answer and allowing them to save the response for future reference.

Key Relationships:

Included Relationships are shown where one use case is part of another, such as "verify username and password" being included in the "login" process.

Extended Relationships are used where a use case might have an additional behavior based on specific conditions, such as the "Display error login" being an extended action during the login process if the credentials are invalid. This diagram provides an overview of how the user interacts with different features of the News Nexus system, and how the system manages these requests, ensuring a seamless experience for accessing news content.

2.1.0.1 High level Use Cases

Use Case	Sign Up
Actor	User
Type	Primary
Description	The user signs up for the system by providing required information like email, username, and password. The system will store the provided data and create a new user account. Once the sign-up is successful, the user can log in and access the system.

Table 2.1: Use Case: Sign Up

Use Case	Logging in
Actor	User
Type	Primary
Description	The user wants to log into the system by entering their username and password. The system will check if the provided credentials are valid. If they are correct, the user will be successfully logged in and can access the system's features. If the login attempt fails, the system will display an error message asking the user to retry.

Table 2.2: Use Case: Logging in

Use Case	Selecting Preferences
Actor	User
Type	Primary
Description	The user selects their news preferences, such as topics they are interested in (sports, politics, health, etc.). The system uses these preferences to personalize the user's news feed and display articles that match their selected topics.

Table 2.3: Use Case: Selecting Preferences

Use Case	Selecting Sources
Actor	User
Type	Primary
Description	The user chooses their preferred news sources, and the system will use those sources to fetch articles. This allows the user to see news only from the sources they trust or prefer.

Table 2.4: Use Case: Selecting Sources

Use Case	Review History
Actor	User
Type	Primary
Description	The user can review their previous interactions or saved responses. The system displays a list of past queries and responses, allowing the user to revisit useful information from past sessions.

Table 2.5: Use Case: Review History

Use Case	Typing Queries
Actor	User
Type	Primary
Description	The user types the query to get the information they wish to have.

Table 2.6: Use Case: Typing Queries

Use Case	Getting an Answer
Actor	User (Primary), System (Secondary)
Type	Primary
Description	The user submits a question or query, either by typing or recording their voice. The system processes the query, retrieves relevant information, and then provides an appropriate response to the user. The interaction is initiated by the user, but the system performs the backend work to fetch and deliver the answer.

Table 2.7: Use Case: Getting an Answer

Use Case	Saving Responses
Actor	User
Type	Primary
Description	After the user gets an answer, they save the responses according to their need, to use it for further use or analysis.

Table 2.8: Use Case: Saving Responses

Use Case	Logging Out
Actor	User
Type	Primary
Description	The user logs out after completing their session.

Table 2.9: Use Case: Logging Out

Expanded Use Case

Use Case	Sign Up
Actor	User
Type	Primary
Preconditions	- User is not registered in the system.
Postconditions	- New user account is created. - User can log in using credentials.
Main Flow	1. User navigates to "Sign Up" page. 2. User enters email, username, and password. 3. Submits the form. 4. System validates details. 5. Account created. 6. Success message displayed.
Alternative Flows	A1: Email Already Registered - Error: "Email already in use." A2: Username Already Taken - Error: "Username unavailable." A3: Invalid Input Data - Invalid details entered. A4: System Error During Registration - Error: "System error." Try again later.

Table 2.10: Use Case: Sign Up

Use Case	Logging In
Actor	User
Type	Primary
Preconditions	- User has an existing account. - System is operational.
Postconditions	- User is authenticated. - User gains access to system features.
Main Flow	1. User navigates to "Login" page. 2. Enters username and password. 3. Clicks "Login". 4. System verifies credentials. 5. User is logged in and redirected to main page.
Alternative Flows	A1: Incorrect Credentials - Error: "Invalid username or password." A2: Account Locked - Error: "Account locked due to multiple attempts." A3: Forgot Password - User initiates password recovery. A4: System Error During Login - Error: "System error. Try again later."

Table 2.11: Use Case: Logging In

Use Case	Selecting Preferences
Actor	User
Type	Primary
Preconditions	- User is logged in. - System has news topics available for selection.
Postconditions	- User's preferences saved. - News feed is updated with selected topics.
Main Flow	1. User navigates to "Preferences" page. 2. User selects topics. 3. Clicks "Save". 4. System updates profile.
Alternative Flows	A1: No Topics Selected - System prompts user to select a topic. A2: System Error While Saving Preferences - Error: "Unable to save preferences at this time."

Table 2.12: Use Case: Selecting Preferences

Use Case	Selecting Sources
Actor	User
Type	Primary
Preconditions	- User is logged in. - System has available news sources.
Postconditions	- User's preferred sources are saved. - News feed is updated accordingly.
Main Flow	1. User navigates to "Sources". 2. Selects preferred sources. 3. Clicks "Save". 4. System updates profile and displays success message.
Alternative Flows	A1: No Sources Selected - User is warned to select sources. A2: Source Unavailable - System notifies the user of unavailable source. A3: System Error While Saving - Error: "Unable to save sources at this time."

Table 2.13: Use Case: Selecting Sources

Use Case	Review History
Actor	User
Type	Primary
Preconditions	- User is logged in. - User has previous interactions or saved responses.
Postconditions	- User reviews previous queries. - User can manage saved responses.
Main Flow	1. User navigates to "History" pane. 2. System retrieves past queries. 3. Displays list of interactions. 4. User selects and views details.
Alternative Flows	A1: No History Available - System shows "No history available." A2: Deleting History Items - User deletes an item from history. A3: System Error Retrieving History - Error: "Unable to retrieve history."

Table 2.14: Use Case: Review History

Use Case	Typing Queries
Actor	User
Type	Primary
Preconditions	- User is logged in. - System is ready to accept queries.
Postconditions	- User's query is submitted. - System proceeds to generate an answer.
Main Flow	1. User navigates to query input area. 2. Types query. 3. Submits query. 4. System processes the query.
Alternative Flows	A1: Empty Query Submission - User is prompted to enter a query. A2: Invalid Characters in Query - System shows error message. A3: System Error Accepting Query - Error: "Unable to submit query."

Table 2.15: Use Case: Typing Queries

Use Case	Getting an Answer
Actor	User (Primary), System (Secondary)
Type	Primary
Preconditions	- User is logged in. - User has submitted a valid query.
Postconditions	- User receives an answer. - System may log the interaction.
Main Flow	1. System receives the query. 2. Processes the query. 3. Generates response. 4. Presents answer to user.
Alternative Flows	A1: Unable to Process Query - System asks the user to rephrase. A2: No Information Found - System informs the user. A3: System Error During Processing - Error: "An error occurred."

Table 2.16: Use Case: Getting an Answer

Use Case	Saving Responses
Actor	User
Type	Primary
Preconditions	- User is logged in. - User has received a response from the system.
Postconditions	- Response is saved to the user's account.
Main Flow	1. When User gets a response, they can save it. 2. Clicks "Save" or "Bookmark". 3. System saves the response.
Alternative Flows	A1: Response Already Saved - System notifies user. A2: Storage Limit Reached - Error: "Reached saved responses limit." A3: System Error While Saving - Error: "Unable to save response."

Table 2.17: Use Case: Saving Responses

Use Case	Logging Out
Actor	User
Type	Primary
Preconditions	- User is logged in.
Postconditions	- User is logged out. - User session is terminated.
Main Flow	1. User clicks on "Logout". 2. System terminates session. 3. Redirects to login page. 4. Displays logout success message.
Alternative Flows	A1: Session Timeout - System automatically logs the user out. A2: System Error During Logout - Error: "Unable to log out."

Table 2.18: Use Case: Logging Out

2.2 Functional Requirements

This section describes the functional requirements of the system expressed in the natural language style. This section is typically organized by feature as a system feature name and specific functional requirements associated with this feature. It is just one possible way to arrange them. Other organizational options include arranging functional requirements by use case, process flow, mode of operation, user class, stimulus, and response depend on what kind of technique has been used to understand functional requirements. Hierarchical combinations of these elements are also possible, such as use cases within user classes.

2.2.1 Module 1: Personalization Module

This module focuses on delivering a personalized news experience for each user.

Requirements:

- **FR1:** The system shall tailor the news feed based on the user's selected preferences (e.g., sports, politics, health) and past interactions.
- **FR2:** The system shall continuously learn from user behavior and interactions to refine and improve the recommendations.

- **FR3:** The system shall create and maintain a user profile based on reading habits, which will be used to improve content relevance.

2.2.2 Module 2: News Content Management Module

This module handles the real-time updates, article summaries, and categorization of news.

Requirements:

- **FR1:** The system shall provide real-time news updates to keep users informed about the latest developments.
- **FR2:** The system shall generate concise article summaries for users to grasp the main points without reading the entire article.
- **FR3:** The system shall categorize news articles into various topics (e.g., politics, sports, technology) to help users navigate through the content.

2.2.3 Module 3: Source Management Module

This module allows users to select and manage their preferred news sources.

Requirements:

- **FR1:** The system shall allow users to choose and filter news based on their preferred news sources.
- **FR2:** The system shall provide an option to modify source preferences at any time.

2.2.4 Module 4: Search and Query Module

This module enables users to search for news articles and handle their queries.

Requirements:

- **FR1:** The system shall allow users to search for news articles or topics using text-based queries.
- **FR2:** The system shall provide a voice search feature to let users search using audio queries.
- **FR3:** The system shall generate video responses for queries related to videos, showing relevant highlights.

2.3 Non-Functional Requirements

2.3.1 Reliability

- Realtime news updates will be made after every 3 hour in to ensure timely access to information. 3 hour time is considered as the news are not available after every sec, so to save the resources and to make content available for users timely, we have used 3 hour time to get up-to-date information.

2.3.2 Availability

- The newsNexus platform should be accessible 24/7, ensuring users can access news and features at any time.
- By standard news data after 15 days is removed from the website so our News Nexus provide previous data effectively and efficiently.

2.3.3 Usability

- The user interface should be intuitive and easy to navigate, allowing new users to familiarize themselves with the platform within 5 minutes.
- The system should provide article summaries, audio integration, and video query responses to cater to users with different preferences and accessibility needs.
- The platform should offer a seamless experience across different devices (desktop, tablet, and mobile) with consistent performance and layout.

2.3.4 Performance

- The platform should load the main news feed quickly to ensure a smooth user experience.
- Search queries should return relevant results within 3 seconds, regardless of the number of articles in the database.
- The personalized news feed should adapt to user preferences within 1 minute of new interaction data being collected.

2.3.5 Security

After developing our front end, according to the situation we will see how to provide security but we plan to secure following for our users: - User data, including preferences and saved articles to ensure privacy and security.

- The system should have multi-factor authentication (MFA) for user accounts to prevent unauthorized access.

2.3.6 Scalability

- When we develop our front end, we will see and work on how the platform should be capable of handling many concurrent users without a significant drop in performance.

- The system should support the addition of new news sources, categories, and features without requiring significant changes to the core architecture.

2.4 Domain Model

The Domain model for my project is below which illustrates the **conceptual model** of the system. The "User" selects preferences and news sources, which are used to display content through the "Interface." The "Interface" shows "History", "News Articles" and allows users to interact with a "Chatbot." Chat interactions can be saved as "Saved Chat" and viewed in the "History." Each component represents different aspects of user interaction, data storage, and content display in the system.

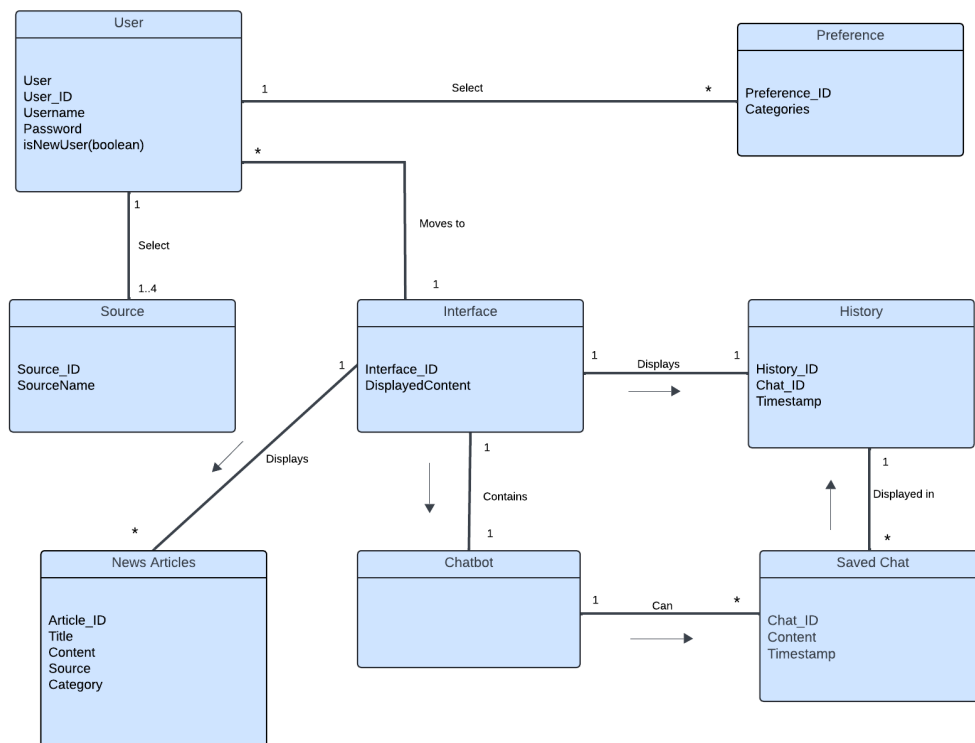


Figure 2.2: Domain Model of News Nexus

Chapter 3

System Overview

"newsNexus" is a smart news aggregation system that's designed to give users a highly personalized and engaging experience when it comes to staying informed. Think of it as your personal news assistant—it brings together articles, real-time video insights, and predictive forecasts, all customized to your preferences.

3.0.1 What It Does

The main idea behind "newsNexus" is to help **you cut through the noise**. There's a lot of news out there, and it can be overwhelming to find what's relevant to you. "newsNexus" simplifies this by continuously collecting articles from a variety of reputable sources, processing them to understand the meaning, and then letting you find exactly what you're interested in through an easy-to-use interface.

You can search using text or even voice commands, and the system will present results based on what it knows you care about. Whether you're interested in traditional articles, want to see what people are saying on YouTube, or are curious about upcoming trends and predictions, "newsNexus" has you covered.

3.0.2 Why It Matters

We live in an age of constant information bombardment—often too much, and not always trustworthy. With "newsNexus", you're in control. You decide the sources you trust and the topics you care about, and the system tailors what you see accordingly. It's designed to make sure you're getting the most accurate and up-to-date information in the way that works best for you.

The system is also smart enough to bring in different forms of media, including articles and videos, so you can get a well-rounded perspective on current events. Plus, with AI-driven

predictive insights, you can get a glimpse into potential future trends for topics that interest you.

3.0.3 How It's Built

Below are the steps to easily understand how it will be built:

1. **Collecting Data:** First, the system goes out and scrapes news from different sites(samaa, ary, geo, 24news, dunya news). This happens regularly to make sure everything is fresh. Once the raw data is collected, preprocessing is done to filter out irrelevant information. The preprocessing includes cleaning the text, extracting useful information, and structuring the data to make it uniform. For further processing only cleaned data is used.
- **Processing Data:** Once the data is collected, it's turned into something the system can understand better—vector embeddings. These are smart representations of the content, making it easier to match what's being asked to what's available. The data is stored in two places—a Vector Database for efficient content retrieval and a SQL Database for the structured details.
- **Processing Query:** When you interact with newsNexus, this is where the magic happens. The system figures out what type of query you're making—basic text searches, questions involving YouTube videos, or future predictions. Depending on what you're asking, it either looks up content in the vector database, processes video clips, or uses AI to generate predictions.
- **Front-End:** The user interface is designed to be clean and intuitive. It lets you log in, set your preferences, search for content, and save your interactions for later. Everything comes together here to give you the news experience you want.
- **Cloud Integration :** Behind the scenes, everything will be hosted in the cloud, which helps the system to handle large amounts of data and serve a lot of users at once. It keeps the system scalable and reliable.
- **Generation Response:** After the system processes your query, this layer is responsible for putting together all the results—from articles, videos, and AI insights—into one comprehensive response, which then gets displayed to you.

3.0.4 Background Inspiration

newsNexus is being developed to solve a real problem. People are overwhelmed by the sheer volume of news and information out there, and it's getting harder to separate valuable, trusted content from everything else. The goal is to build a tool that feels like it's working for you, curating what's important based on your interests and providing a complete view of each topic.

We also wanted to make it future-oriented. By including AI-based predictions, we help users not just understand what's happening now, but also get an idea of what might happen next. And by adding the YouTube feature, newsNexus lets users engage with trending topics in a more dynamic way.

Overall, newsNexus is about giving people a personalized, in-depth news experience that's easy to navigate and always up to date.

3.1 Architectural Design

The architecture is designed to be modular and efficient, ensuring that user queries are handled swiftly and the right information is served back to the user in a meaningful way. Each layer plays a distinct role, and together they form a robust pipeline from data collection to content delivery.

1. Front-End Layer The front-end layer serves as the interface between the user and the system, providing a user-friendly and intuitive experience. It enable users to interact with the system by submitting queries, viewing results, and accessing their preferences. The focus here is on creating a seamless experience for users to quickly and easily find the news content or other information they need. This layer is responsible for gathering input from the user (like text or audio queries) and displaying organized results from various sources.

2. Query Processing Layer This is the heart of the system where the magic happens. Once a query is submitted by a user, this layer is responsible for classifying the type of query. The query can be categorized as a Basic Query, a YouTube Query, or a Future Prediction Query.

- **Basic Query:** For text-based searches or direct information retrieval.
- **YouTube Query:** If the user wants to find specific content from YouTube, this module processes videos to extract relevant segments.
- **Future Prediction Query:** When a user asks about a trend or forecast, this module uses an AI/ML Model to generate predictive insights.

Depending on the type of query, this layer interacts with various modules like the AI model or video processing module to get the relevant content. It then aggregates the results and passes them to the front-end for display.

3. Data Processing Layer The data processing layer ensures that all the data is in the right form for efficient retrieval and analysis. It has two major components:

- **Vector Database:** Stores vector embeddings that are derived from text data. This helps in quickly matching and retrieving relevant content based on similarity.
- **SQL Database:** Contains structured data like the cleaned, preprocessed text, saved responses and metadata.

This layer is responsible for generating these vector embeddings, which transform text into a format suitable for quick retrieval and comparison. Once the data is processed, it is stored in a format that allows fast access and effective searching, making the system efficient and responsive.

4. Data Collection Layer The data collection layer is where raw data comes into the system. It handles the process of web scraping, where news and information from various sources are collected periodically. After the data is scraped, it undergoes preprocessing to clean, standardize, and extract relevant information before it is handed over to the data processing layer. This ensures that only the most accurate and useful data moves forward in the pipeline. The entire process is automated to keep the system updated with fresh content regularly.

5. Cloud Integration Layer This layer supports the entire architecture by utilizing cloud services that gives scalability, ensuring that as more data is collected or more users access the system, the system can handle the increased load smoothly.

6. Response Generation Layer This layer acts as a bridge between the query processing and front-end layers. After a query is classified, processed, and relevant content is retrieved, this layer aggregates all the pieces of information (e.g., text-based articles, videos, predictions) into a coherent response. It ensures that the final output is organized, easy to understand, and ready for presentation on the front-end for the user.

3.1.1 High-Level Overview of Architectural Design

The newsNexus system efficiently fetches, processes, and displays news content in a user-friendly way. Here's how all modules work together to achieve this:

1-Data Collection to Storage:

The process starts with the Data Collection Layer, where the Web Scraping Module gathers news from various sources. The data is then preprocessed, cleaned, structured, and transformed into vector embeddings for efficient retrieval. The processed data is stored in:

- **Vector Database:** For content retrieval based on semantic meaning.
- **SQL Database:** For structured data and metadata.

2-User Interaction and Query Handling:

Users interact with the system via the Front-End Layer to search, update preferences, or change sources. The Query Processing Layer classifies the user's query as:

- **Basic Query:** Direct content retrieval.
- **YouTube Query:** Searches for and processes relevant video clips.
- **Future Prediction Query:** Uses AI/ML to predict outcomes based on data.

3-Fetching and Generating Responses:

The Query Processing Layer interacts with the Data Processing Layer to fetch articles, process YouTube videos, or generate predictions. The Response Generation Layer then aggregates all the results into a comprehensive response.

4-Personalization and Cloud Integration:

User preferences are considered throughout the query processing for personalized content. The architecture is hosted in the Cloud Integration Layer, ensuring scalability, storage, and seamless communication.

5-Displaying Results to User:

The Response Generation Layer sends the aggregated content back to the Front-End Layer for display. This enables users to interact with content types and receive highly relevant, personalized results in real time.

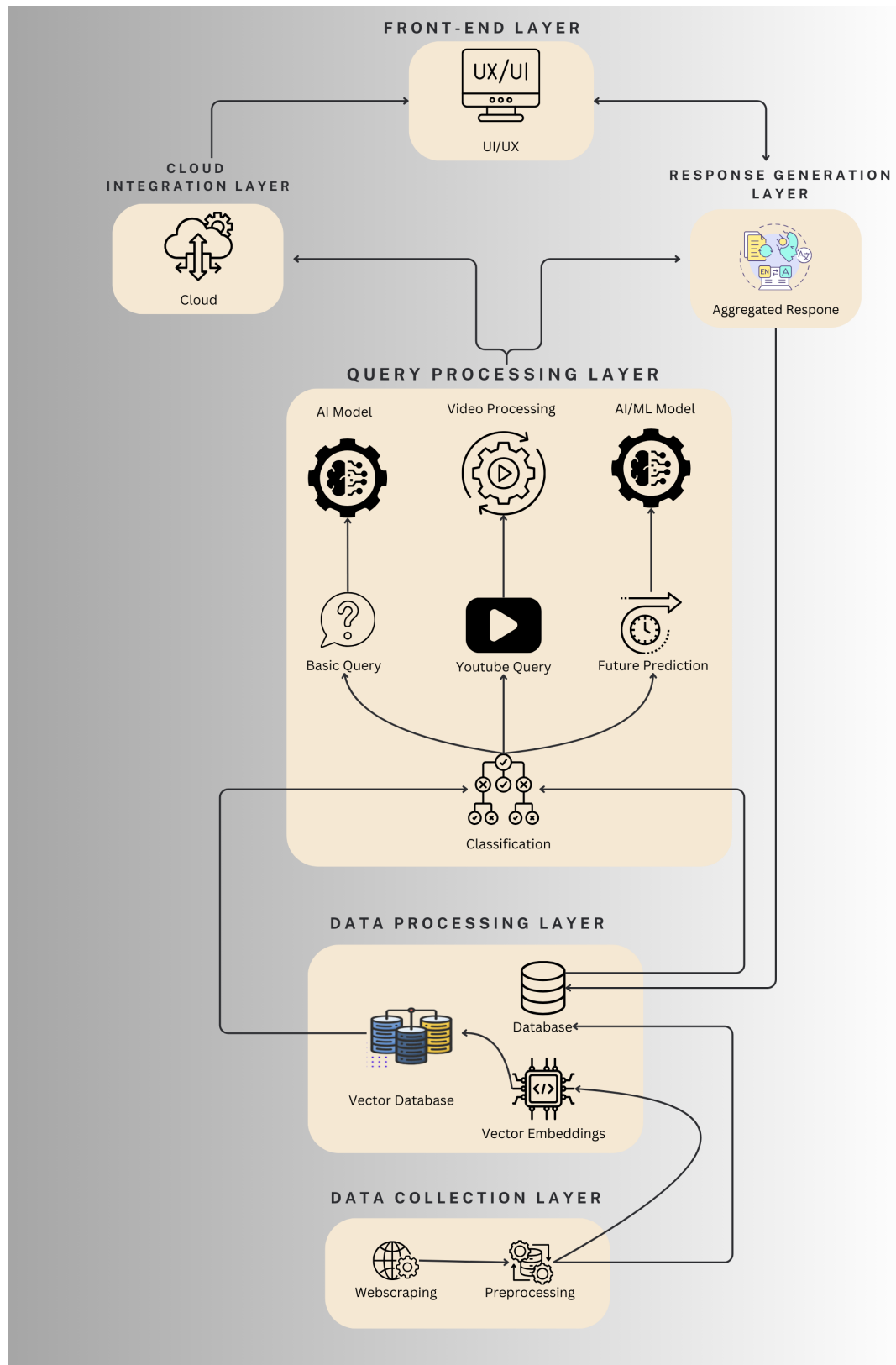


Figure 3.1: Architectural Design of News Nexus

3.2 Design Models

3.2.1 System Sequence Diagram

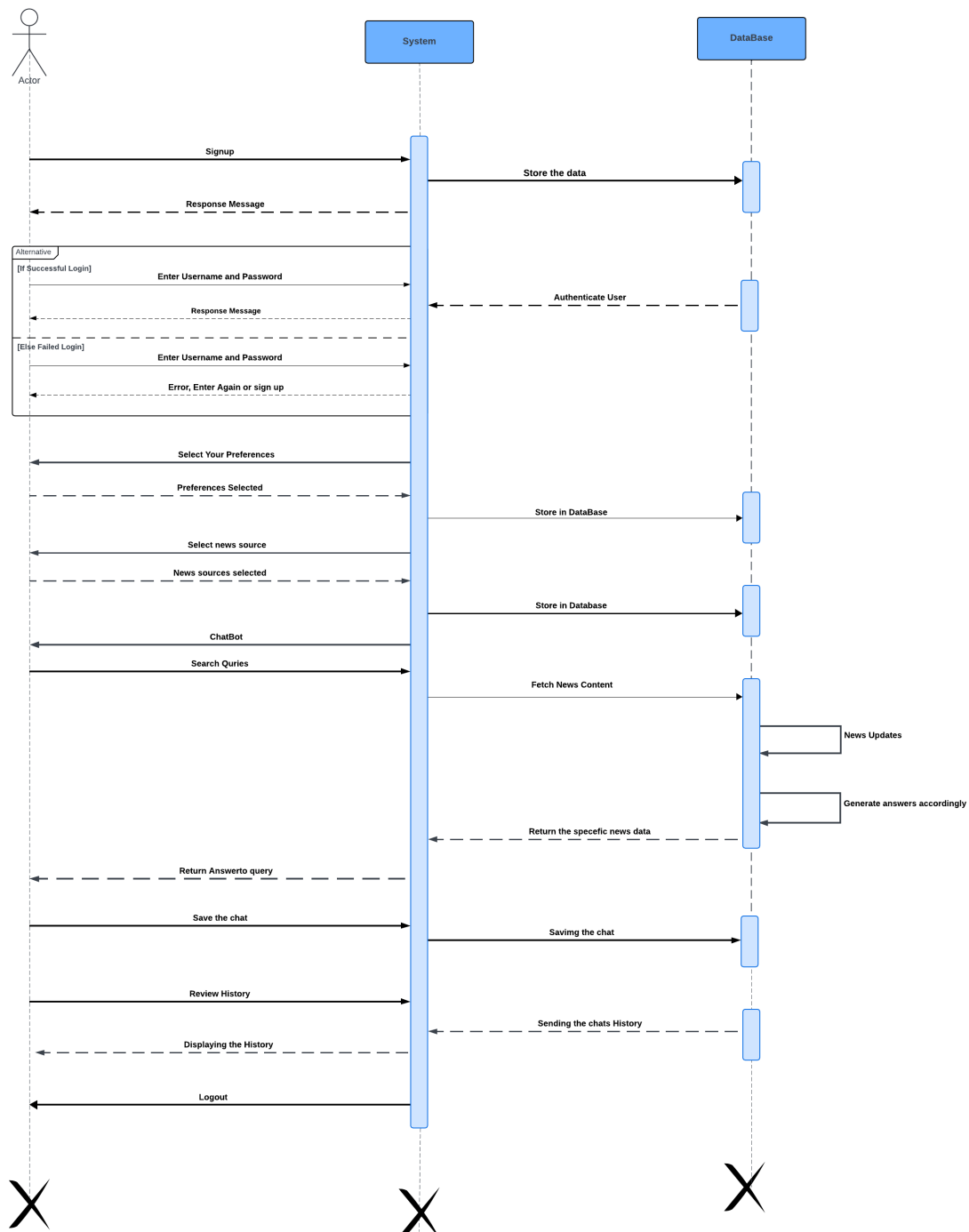


Figure 3.2: System Sequence Diagram of News Nexus

The above sequence diagram shows the flow of user interactions with the system.

3.2.2 Activity Diagram

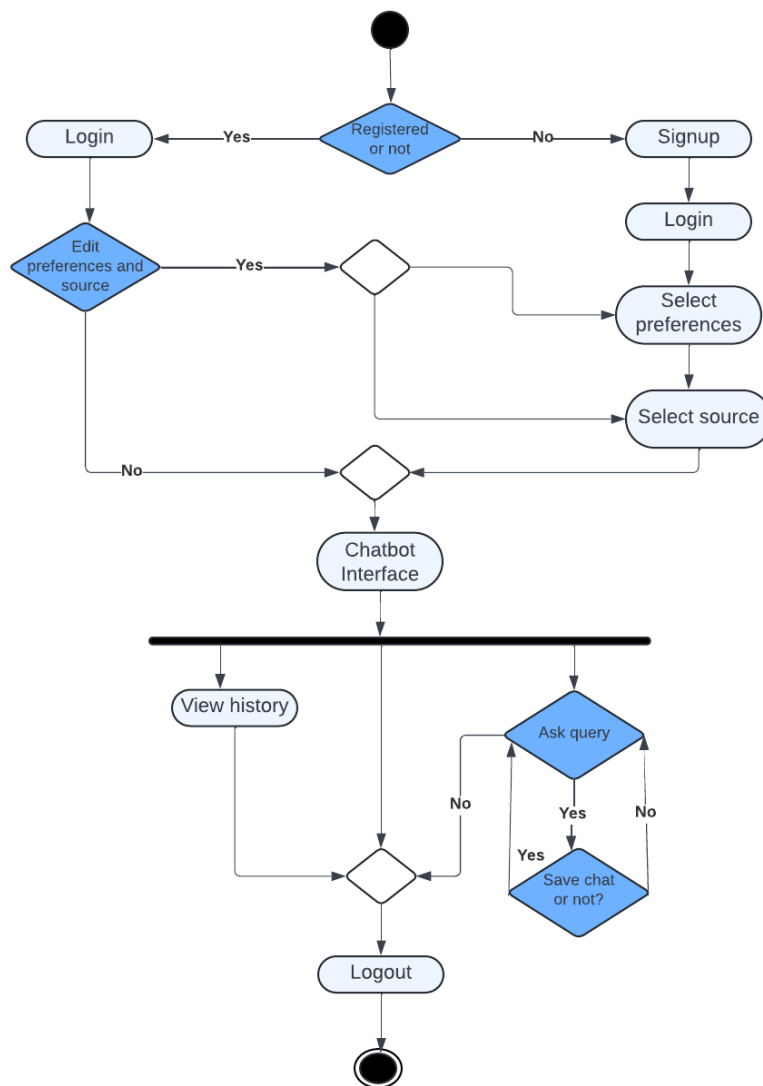


Figure 3.3: Activity Diagram of News Nexus

This activity diagram illustrates the flow of user interaction within "newsNexus", from registration to query handling and logout.

- The process begins by checking if the user is already registered. If not, the user is directed

to sign up before proceeding.

- Once registered, the user can log in and optionally edit their preferences and news sources.
- After setting their preferences, the user can access the chatbot interface.
- In the chatbot interface, the user can view their interaction history or ask new questions.
- If a new query is submitted, the user has the option to save the chat before concluding.
- The activity diagram also shows the ability for the user to log out, ending their session.

The diagram captures the user's journey and decision points throughout the system, ensuring an easy-to-follow, interaction flow.

3.2.3 Data Flow Diagram

Level 0

This level 0 data flow diagram provides an overview of the entire system, depicting key process, data flow between users, and major external entity.

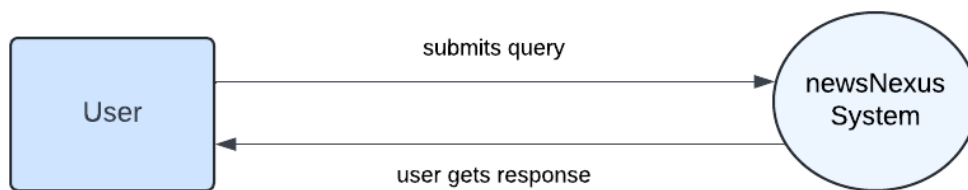


Figure 3.4: level 0 - Data Flow Diagram

Level 1

This Level 1 Data Flow Diagram shows the flow of data between the user, query classification, processing and external data sources. The user submits a query, after cleaning it is processed by the "Classify Query". This component interacts with two databases: an SQL database for structured data and a Vector database for embeddings. Additionally, the query may trigger a YouTube API search for video content, providing relevant video data back to the system. Once all necessary data is collected and processed, a response is generated and sent back to the user, completing the data flow cycle.

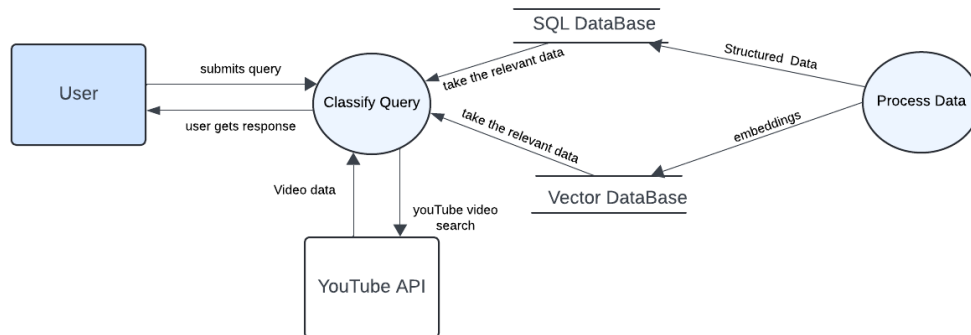


Figure 3.5: level 1 - Data Flow Diagram

Level 2

• Level 2.1- Diagram for Processing Data Process

This Level 2 Data Flow Diagram shows the process of scraping data, preprocessing it, and generating vector embeddings. First, data is scraped and then sent to the "Preprocess Data" step to clean it and transform it into a structured form. This structured data is stored in an SQL database, while the clean data is also used to generate vector embeddings, which are stored in a vector database. The diagram depicts how the system interacts with databases to manage structured data and embeddings.

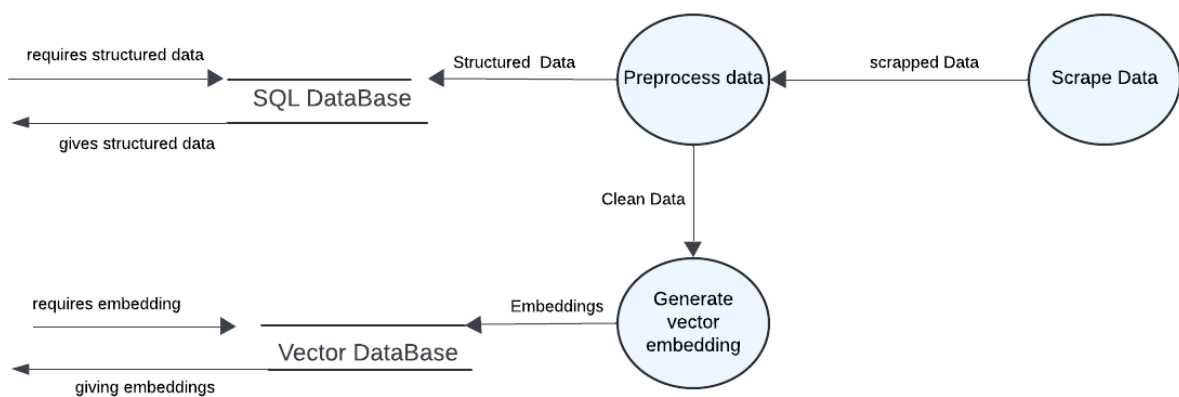


Figure 3.6: level 2.1 - (Process Data) Data Flow Diagram

• Level 2.2- Diagram for Classify Query Process

This Level 2.2 Data Flow Diagram shows how user queries are classified and processed. When a user submits a query, it is sent to the "Classify and Process Query" module, which routes it based on its type—General Query, Prediction Query, or YouTube Query. General queries fetch relevant data, prediction queries involve retrieving embeddings from the SQL/Vector Database, and YouTube queries utilize the YouTube API to gather video data. Once the appropriate data is gathered, the "Compile Response" module processes and sends the final response back to the user.

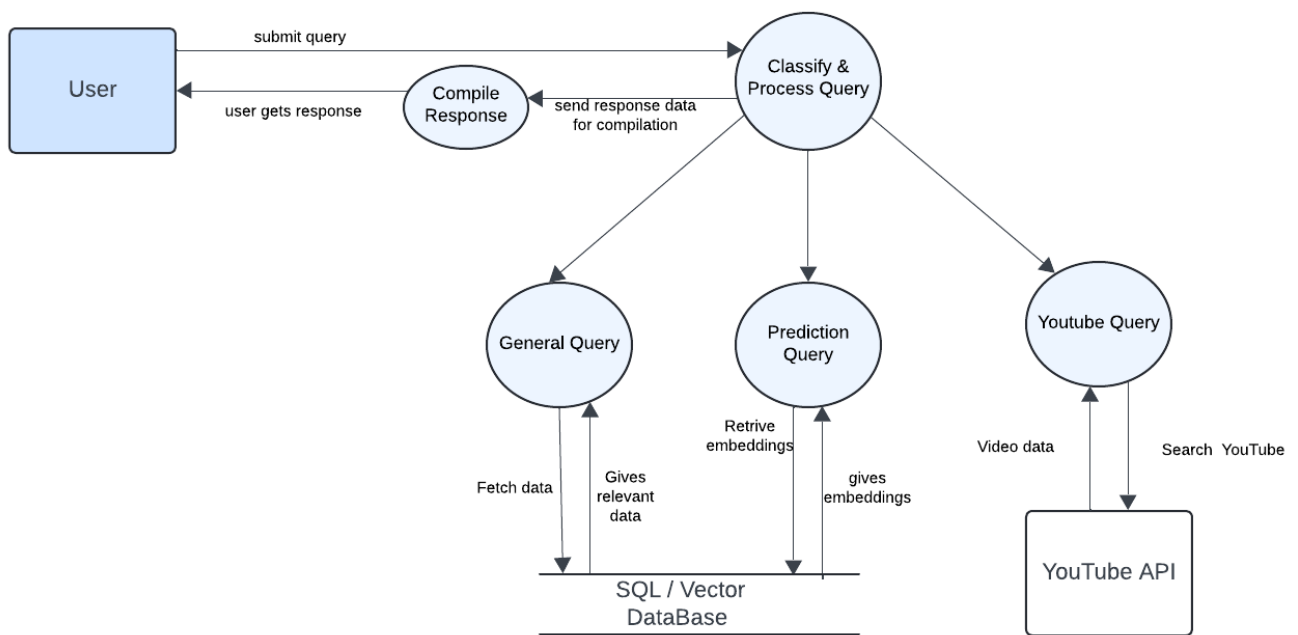


Figure 3.7: level 2.2 - (Classify Query) Data Flow Diagram

3.3 Data Design

The data collection process starts by scraping news articles from five different sources, which are saved into their respective CSV files. Each source provides various details like the title, author, post date, and more. After the data is collected, we focus on extracting the specific columns we need—such as category, title, author, and URL—so we only keep the important information. Once this step is done, the filtered data from all the sources is combined into

a unified format. This combined data is then saved into a database, as shown in the image, which is used in the next stages of the project. The image below shows an overview of data collection :

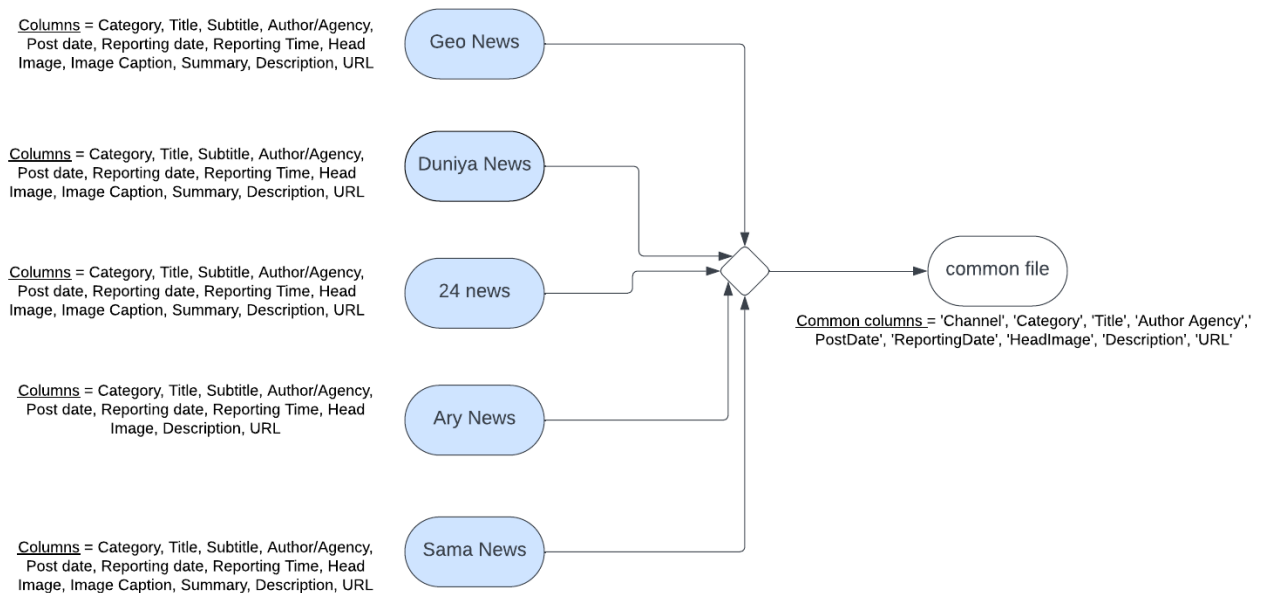


Figure 3.8: Data Collection Flow

Below is the combined data of all the csv in the database:

SQLQuery1.sql - not connected*

```
use newsData;
select * from newsArticles;
```

133 %

	Channel	Category	Title	AuthorAgency	PostDate	ReportingDate	ReportingTime	HeadImage	Description	URL
1	Any News	pakistan	Sindh being subjected to injus...	Web Desk	2024-07-06	2024-07-06	16:32:38	https://arynews.tv/wp-content...	JAMSHORO: Sindh being subjected to injus...	https://arynews.tv/sindh-being-subject...
2	Any News	pakistan	WAPDA generates 34.4b...	Web Desk	2024-07-06	2024-07-06	16:32:48	https://arynews.tv/wp-content...	ISLAMABAD: As many as 22 Hydel Power...	https://arynews.tv/wapda-generates-3...
3	Any News	pakistan	PTI Islamabad power show...	Web Desk	2024-07-06	2024-07-06	16:33:33	https://arynews.tv/wp-content...	ISLAMABAD: Pakistan Tehreek-e-Insaf (PT...	https://arynews.tv/pti-islamabad-powe...
4	Any News	international	US Dollar on defensive aft...	Reuters	2024-07-04	2024-07-06	16:52:53	https://arynews.tv/wp-content...	The pound and the euro gained against US...	https://arynews.tv/dollar-rate-today-int...
5	Geo News	World	92 News shuts down in UK...	Not Found	2024-08-04	2024-08-06	14:07:34	https://www.geo.tv/assets/upl...	LAHORE: Urdu-language TV channel 92 Ne...	https://www.geo.tv/latest/557495-92-r...
6	Geo News	Amazing	Breaching whale capsizes...	Not Found	2024-07-25	2024-08-06	14:07:43	https://www.geo.tv/assets/upl...	A breaching whale, in the United States Tue...	https://www.geo.tv/latest/555854-bre...
7	Any News	pakistan	Monsoon currents to enter...	Web Desk	2024-07-06	2024-07-06	16:34:57	https://arynews.tv/wp-content...	KARACHI: The monsoon currents are expe...	https://arynews.tv/monsoon-currents-i...
8	Any News	pakistan	Security beefed up around...	Web Desk	2024-07-06	2024-07-06	16:35:09	https://arynews.tv/wp-content...	LAHORE: In preparation for Muharram, DIG...	https://arynews.tv/security-beefed-up...
9	Any News	pakistan	Top military brass says Op...	Web Desk	2024-07-06	2024-07-06	16:37:05	https://arynews.tv/wp-content...	ISLAMABAD: General Syed Asim Munir, NI...	https://arynews.tv/pakistan-corps-corr...
10	Any News	pakistan	Punjab seeks Army, Range...	Web Desk	2024-07-05	2024-07-06	16:37:42	https://arynews.tv/wp-content...	LAHORE: The Punjab government has sou...	https://arynews.tv/punjab-seeks-army...
11	Any News	pakistan	Utility Stores employees ar...	Web Desk	2024-07-05	2024-07-06	16:39:33	https://arynews.tv/wp-content...	In a shocking incident, two staff members of...	https://arynews.tv/utility-stores-staff-a...
12	Any News	pakistan	Bitawal Bhutto admits elect...	Web Desk	2024-07-05	2024-07-06	16:40:38	https://arynews.tv/wp-content...	QUETTA: Pakistan People's Party (PPP) Ch...	https://arynews.tv/bitawal-bhutto-admi...
13	Any News	international	Hamas agrees on talks to f...	Reuters	2024-07-06	2024-07-06	16:42:40	https://arynews.tv/wp-content...	Hamas has accepted a U.S. proposal to be...	https://arynews.tv/hamas-agrees-on-t...
14	Any News	international	'Bridgerton' actor Nicola Co...	Web Desk	2024-07-06	2024-07-06	16:43:45	https://arynews.tv/wp-content...	"Bridgerton" actor Nicola Coughlan has rais...	https://arynews.tv/bridgerton-actor-nic...
15	Any News	international	Bitcoin set for worst week i...	Reuters	2024-07-05	2024-07-06	16:44:58	https://arynews.tv/wp-content...	Bitcoin was set for its biggest weekly fall in ...	https://arynews.tv/bitcoin-rate-today-j...
16	Any News	international	Record number of UK Cabi...	AFP	2024-07-05	2024-07-06	16:46:30	https://arynews.tv/wp-content...	A record number of Cabinet ministers lost th...	https://arynews.tv/record-number-of-u...
17	Any News	international	Biden says 'I'm not going a...	Reuters	2024-07-05	2024-07-06	16:47:09	https://arynews.tv/wp-content...	President Joe Biden said "I'm not going any...	https://arynews.tv/biden-says-im-not-g...
18	Any News	international	Polls open in Iran presid...	AFP	2024-07-05	2024-07-06	16:47:22	https://arynews.tv/wp-content...	Polls opened Friday for Iran's runoff preside...	https://arynews.tv/polls-open-in-iran-p...
19	Any News	international	UK general election 2024...	Reuters	2024-07-05	2024-07-06	16:48:17	https://arynews.tv/wp-content...	Labour surged to a landslide victory in UK's ...	https://arynews.tv/uk-general-election...
20	Any News	international	Copa America Empty seat...	Reuters	2024-07-04	2024-07-06	16:49:32	https://arynews.tv/wp-content...	As the Copa America 2024 enters the knoc...	https://arynews.tv/copa-america-2024...
21	Any News	international	Hezbollah rains rockets on...	AFP	2024-07-04	2024-07-06	16:51:43	https://arynews.tv/wp-content...	BEIRUT: Lebanon's Hezbollah said it launc...	https://arynews.tv/hezbollah-rains-roc...

Disconnected.

Ln 1 Col 4 INS

Figure 3.9: Combined Data

Chapter 4

Conclusion and References

4.1 Conclusion

By the time we reached our mid-evaluation, we have successfully completed several important milestones for our project. We started by thoroughly gathering requirements, making sure we understood the project goals and what our users needed. After that, we focused on implementing web scraping to collect data from various sources, which we then cleaned and organized during the preprocessing stage.

We also set up SQL and vector databases to store the data efficiently and make it easy to retrieve when needed. Finally, we developed a general query and response system that allows users to interact with the data and get meaningful answers. These foundational achievements have put us in a great position for the next phase, where we'll refine these components further and build a user-friendly, robust system.

4.2 References

Below are the links of the news channels we scrapped:

- <https://www.24newshd.tv/>
- <https://www.24newshd.tv/>
- <https://arynews.tv/>
- <https://www.samaa.tv/>
- <https://www.geo.tv/>

To better understand the concepts behind the diagrams made, we explored the following resources:

- <https://www.edrawsoft.com/article/uml-system-sequence-diagram.html>
- <https://miro.com/diagramming/what-is-software-architecture-diagramming/>
- <https://www.thoughtworks.com/insights/blog/agile-project-management/domain-modeling-what-you-ne>
- <https://venngage.com/blog/data-flow-diagram/>